

What is the effect of the Korea-US Free Trade Agreement on Global Value Chains?

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August 6, 2026

Abstract

This paper empirically examines the effect of the Korea–U.S. (KORUS) Free Trade Agreement (FTA) on global value chain (GVC) trade and non-GVC trade between Korea and the United States. This distinction is important because international trade has increasingly shifted from trade in final goods to trade in value-added inputs along global value chains. Using synthetic controls, this paper estimates the treatment effect of the FTA on gross exports, GVC trade, and non-GVC trade in three sectors: autos, agriculture, and electronics. U.S. auto imports from Korea increased by an average of \$4.8 billion, while U.S. agricultural exports to Korea declined by \$0.7 billion. U.S. auto and agricultural exports to Korea also expanded substantially, accompanied by rising GVC intensity. The FTA had minimal effects on the electronics sector. While the sectoral results are mixed, the overall pattern suggests that the FTA led to a bilateral restructuring of GVCs, with Korea exporting more finished goods directly to U.S. consumers and the United States supplying more intermediates and components for Korean production. A robustness check using a gravity model yields slightly different results by sector, but supports the same overall conclusion.

JEL CLASSIFICATION: F13, F15, F61

KEY WORDS: International Trade, Free Trade Agreement, Global Value Chain.

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1 Introduction

This paper examines the effect of one major policy change, the Korea-US Free Trade Agreement (KORUS FTA) on global value chain (GVC) trade and non-GVC trade between Korea and the United States. This is an important empirical question for two reasons. First, the KORUS FTA is the second-largest U.S. FTA by trade flows, making it a major and economically important agreement. Second, international trade has increasingly shifted from trade in final goods to trade in value-added inputs along global value chains. Examining how the KORUS FTA affects not only bilateral trade between the two countries but also GVC and non-GVC trade flows provides important insights into how FTAs affect trade patterns in a highly integrated global economy characterized by widespread GVC participation.

Several previous studies on the potential effects of KORUS, summarized in USITC (2001, 2010), estimate that the agreement would significantly increase U.S.–Korea trade, with U.S. exports to Korea rising more than U.S. imports from Korea. Yaylaci and Shikher (2014) report similar findings, with U.S. exports to Korea increasing by 56.9% and Korean exports to the United States increasing by 18.9%. In addition, both countries are projected to further specialize in industries where they have strong technological comparative advantages, while experiencing relatively modest welfare gains. In contrast, this paper finds that, at least in the automotive and agricultural sectors, U.S. imports from Korea increased more than U.S. exports to Korea. Additionally, this paper contributes by examining the effects of KORUS on the restructuring of global value chains in U.S.–Korea trade, an aspect largely overlooked in the literature, which has instead focused on the effects on GDP, productivity, trade flows, trade diversion, and welfare (Lee and Song, 2008; Choi and Schott, 2004; Lee and Lee, 2005; Kiyota and Stern, 2007; Russ and Swenson, 2019).

Using synthetic controls, this paper estimates the treatment effect of the FTA on gross exports, GVC trade, and non-GVC trade in three sectors: autos, agriculture, and electronics. U.S. auto imports from Korea increased by an average of \$4.8 billion, while U.S. agricultural exports to Korea declined by \$0.7 billion. U.S. auto and agricultural exports to Korea also expanded substantially, accompanied by rising GVC intensity. The FTA had minimal effects on the electronics sector. While the sectoral results are mixed, the overall pattern suggests that the FTA led to a bilateral restructuring of GVCs, with Korea exporting more finished goods directly to U.S. consumers and the United States supplying more intermediates and components for Korean production. A robustness check using a gravity

model yields slightly different results by sector, but supports the same overall conclusion. These results suggest that FTAs, depending on their specific provisions and bilateral trade flow, can simplify trade and reduce the importance of GVCs, rather than deepening trade integration and increasing the importance of GVCs.

GVC trade is important because participation in global value chains raises labor productivity through greater production efficiency. By specializing in production stages aligned with their comparative advantage, countries reduce costs and increase their international competitiveness (Constantinescu et al., 2019; Xing et al., 2021). GVC trade plays a particularly critical role for emerging economies by promoting the diffusion of technology and production expertise. By participating in global value chains, developing countries can specialize in individual stages of production rather than producing complete final goods, thereby reducing barriers to industrialization (World Bank, 2019; WTO, 2019).

Free trade agreements (FTAs) can influence GVC trade through two main channels. First, FTAs reduce barriers to trade by lowering tariff rates and mitigating non-tariff barriers through greater customs transparency and streamlining regulatory procedures. Since GVC trade crosses multiple borders, reductions in these barriers can have amplified effects on overall trade flows (Miroudot et al., 2013; Blanchard et al., 2025). Second, FTAs provide a more stable trade environment by establishing binding long-term commitments, which encourage firms to form and maintain long-term production relationships (Baccini et al., 2021; Ruta, 2017; Laget et al., 2020). GVCs rely on predictable and stable trade environments, given that production is fragmented across multiple firms and jurisdictions.

Conversely, FTAs could simply expand non-GVC trade as a direct result of reduced tariffs and non-tariff barriers between the United States and Korea (Cavallo et al., 2021; Amiti et al., 2019). Lower tariffs translate into lower prices, which in turn increase demand for the product (Edwards, 1993; Wacziarg and Welch, 2008; Dornbusch, 1992). In certain instances, when an FTA focuses on liberalizing final goods, value chains that were established to avoid tariff protection may no longer be needed. In such cases, intermediate goods trade that had developed to avoid tariffs may decline after liberalization. Thus, GVC trade can either expand or contract depending on where liberalization occurs within the production chain. This paper contributes to the literature by providing an empirical estimate of how the KORUS FTA has affected GVC and non-GVC trade between the United States and Korea.

To evaluate the impact of the KORUS FTA on GVC trade, this paper employs value-added accounting instead of conventional gross trade measures, which double-count value

added when global value chains are involved (Hummels et al., 2001a; Johnson and Noguera, 2012a; Koopman et al., 2014a; OECD, 2012). Using value-added accounting, gross exports can be decomposed into GVC trade and non-GVC trade. An increase in gross exports may be driven by expansion in non-GVC trade, GVC trade, or both (Borin and Mancini, 2015; Antràs and Chor, 2021). Therefore, the effect of an FTA on GVC trade is ultimately an empirical question.

2 KORUS FTA Institutional Details

Trade between the United States and Korea is asymmetric with the United States running a trade deficit with Korea. Consequently, U.S. negotiators were particularly focused on securing market access in sectors where they believed U.S. exports could expand. After lengthy negotiations and political debates in both countries, the KORUS FTA entered into force in March 2012 (Williams et al., 2014).¹ The U.S. objective in the FTA was to remove barriers to the South Korean auto and agricultural markets. Korea’s agricultural sector was widely regarded as one of the most protected among OECD members, and regulatory standards in the auto industry were viewed as limiting foreign competition. Conversely, Korea pursued greater market openness to enhance national competitiveness and sought expanded access to U.S. markets for automobiles and other manufactured exports (Schott, 2010). Given U.S. priorities, the FTA focused primarily on the agriculture and auto sectors. However, the agreement also reduced and eliminated both tariff and non-tariff barriers and increased regulatory transparency across several sectors (USITC, 2007; Cooper et al., 2011).

2.1 Economic Setting of KORUS FTA

Table 1 lists the top 10 U.S. export and import partners in 2012. South Korea ranks as the 8th largest export destination and the 6th largest import source, accounting for 2.7% of U.S. exports and 2.6% of U.S. imports, respectively. These figures underscore South Korea’s importance as a U.S. trading partner, aside from its NAFTA partners, Canada and Mexico.

¹The agreement was initially signed in 2007, but the U.S. Congress did not ratify its original version. A revised agreement was subsequently signed in 2011.

Table 1: US Export and Import Partners 2012

Rank	Export Partner	Export	Export Share (%)	Import Partner	Import	Import Share (%)
	World	1545	100	World	2274	100
1	Canada	293	18.9	China	426	18.7
2	Mexico	216	14.0	Canada	323	14.2
3	China	111	7.2	Mexico	278	12.2
4	Japan	70	4.5	Japan	146	6.4
5	United Kingdom	55	3.6	Germany	109	4.8
6	Germany	48	3.1	Korea, Rep.	59	2.6
7	Brazil	44	2.8	Saudi Arabia	56	2.5
8	Korea, Rep.	42	2.7	United Kingdom	55	2.4
9	Netherlands	41	2.6	France	42	1.8
10	Hong Kong	37	2.4	India	41	1.8

Export and Imports in billions of \$

Source: WITS

In comparison, Table 2 reports Korea's top 10 export and import partners. As shown in Table 2, the United States is Korea's second-largest export destination and third-largest import source, representing 10.7% of exports and 8.4% of imports. This highlights Korea's stronger trade dependence on the United States. Notably, for both countries, China is the largest import partner.

Table 2: Korea Export and Import Partners

Rank	Export Partner	Export	Export Share (%)	Import Partner	Import	Import Share (%)
	World	548	100	World	520	100
1	China	134	24.5	China	81	15.6
2	United States	59	10.7	Japan	64	12.4
3	Japan	39	7.1	United States	44	8.4
4	Hong Kong	33	6.0	Saudi Arabia	40	7.6
5	Singapore	23	4.2	Qatar	26	4.9
6	Vietnam	16	2.9	Australia	23	4.4
7	Indonesia	14	2.6	Kuwait	18	3.5
8	India	12	2.2	Germany	18	3.4
9	Russia	11	2.0	Indonesia	16	3.0
10	Brazil	10	1.9	UAE	15	2.9

Export and Imports in billions of \$

Source: WITS

Table 3a reports the main U.S. export sectors to Korea. Business activities rank as the largest sector, largely driven by aircraft leasing to Korea, followed by chemicals, electronics, and autos. Table 3b presents the primary U.S. import sectors from Korea, where autos and electronics are among the largest categories. Autos, agriculture, and electronics together comprise roughly 30% of U.S. exports to Korea, while for Korea, these sectors account for over 50% of exports to the United States. The prominence of these sectors played a central role in shaping the KORUS negotiations, reflecting both the United States' position as the dominant partner and Korea's strong export dependence on the U.S. market. For these reasons, the analysis focuses on the auto, agriculture, and electronics sectors.²

Although this paper focuses only on manufacturing sectors, it is worth noting that service sectors are also affected by the FTA and warrant separate analysis. Services trade differs from goods trade in two important ways. First, GVCs involving services encompass a broader range of transactions, including those that occur within countries between domestic and foreign entities, transactions that are not typically captured in GVCs involving goods. Second, the policies affecting services trade are qualitatively different from those affecting goods trade. For example, tariffs are generally less relevant for services trade, whereas regulatory measures play a much larger role and are typically less central in goods trade.

Table 3: Top 10 U.S. Trade Sectors with Korea (2012)

(a) Exports			(b) Imports		
Sector	Exports (USD)	Share (%)	Sector	Imports (USD)	Share (%)
Business Activities	8.25	17.84	Auto	18.27	26.56
Chemicals	6.96	15.06	Electronic	17.73	25.78
Electronic	5.87	12.69	Chemicals	5.57	8.10
Auto	4.87	10.52	Machinery	4.91	7.14
Machinery	4.19	9.05	Petroleum	4.79	6.97
Metals	2.63	5.69	Metals	4.76	6.91
Food	2.55	5.52	Business Activities	4.62	6.72
Agriculture	2.37	5.12	Textiles	2.29	3.33
Paper	1.27	2.74	Rubber	1.22	1.78
Telecom	1.22	2.64	Retail	0.73	1.06

Notes: Values are reported in billions of U.S. dollars. Shares represent each sector's percentage of total bilateral trade in 2012.

²Given that U.S. agricultural imports from Korea are minimal, they are omitted from the analysis.

2.2 Sectoral Provisions of KORUS FTA

This section outlines the specific provisions of the KORUS FTA related to the auto, agriculture, and electronics sectors. In the auto sector, the agreement included measures affecting both GVC and non-GVC trade. It eliminated duties on virtually all auto parts. For U.S. imports from Korea, the 2.5% tariff on passenger vehicles was scheduled for elimination in 2016, while the 25% tariff on light trucks was to be phased out between 2019 and 2022. These provisions were later revised in 2019 under the Trump administration. For U.S. exports to Korea, the 8% tariff on passenger vehicles was reduced to 4% and eliminated in 2016, and the 10% tariff on trucks was removed.

Beyond tariff reductions, the FTA increased regulatory transparency and harmonized safety and environmental standards. It permitted each automaker to self-certify up to 25,000 vehicles annually under U.S. federal safety standards, allowing them to bypass certain Korean requirements. The agreement also reduced Korea's engine displacement tax, which had disproportionately affected larger U.S. vehicles, and established an Automotive Working Group to address ongoing regulatory and safety concerns. These provisions did not apply to the approximately 700,000 Korean vehicles assembled annually in Hyundai and Kia plants located in the United States.

For the agriculture sector, South Korea agreed to grant duty-free access to roughly two-thirds of existing U.S. agricultural exports. This is quite significant because Korea is the fifth-largest overall market for U.S. agricultural products, valued at approximately \$5 billion. Tariffs and import quotas on the remaining products are scheduled to be phased out within 10 years. In addition, South Korea committed to eliminating its 40% tariff on U.S. beef over 15 years and to phasing out its 25% tariff on frozen pork by 2016.

For the electronics sector, the FTA introduced no additional tariff reductions, as the Multilateral Information Technology Agreement had already eliminated tariffs on most electronic products prior to KORUS. However, the agreement strengthened provisions related to non-tariff measures, including intellectual property protection, government procurement, and competition policy.

The results indicate that these provisions had significant effects in the auto and agriculture sectors, but not in electronics. The agreement appears to have temporarily increased U.S. auto exports to Korea while also raising U.S. auto imports from Korea. Interestingly, despite improved market access conditions, U.S. agricultural exports to Korea declined in gross terms, contrary to the agreement's stated objective. This suggests that trade pol-

icy provisions alone were insufficient to overcome domestic resistance in Korea, especially consumer concerns regarding U.S. beef and mad cow disease.

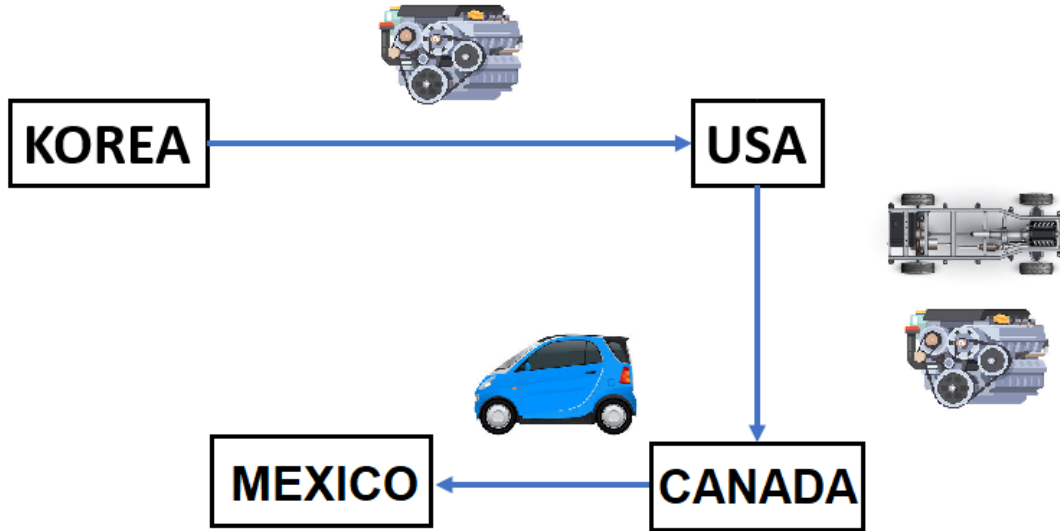
Another aspect of this narrative is tariff escalation, whereby tariffs increase with the level of processing and create incentives that shape trade patterns along the value chain. Lower tariffs on intermediate inputs facilitate cross-border production, while higher tariffs on final goods encourage specialization in different stages of production. In the case of KORUS, tariff reductions, particularly in autos, appear to have weakened the escalation structure on final goods for Korea, enabling it to export more finished vehicles directly to U.S. consumers, while increased access to Korean markets increased the United States' supply of intermediate goods into Korean production.

3 Global Value Chains

3.1 What is GVC trade?

In a global value chain, production is fragmented across multiple countries rather than completed within a single economy. For example, Figure 1 illustrates Korea exporting a car engine to the United States. The United States produces a car chassis, installs the Korean engine, and exports the partially assembled vehicle to Canada. Canada then completes the vehicle and exports the finished car to Mexico. In this example, no single country produces the car entirely. Instead, each specializes in a particular stage of production. The trade flows between Korea, the United States, and Canada therefore constitute GVC trade.

Figure 1: Global Value Chain Diagram



World Bank (2019)

3.2 How do we measure GVC trade?

The measure most commonly used in the literature defines GVC trade as trade in products whose components have crossed at least two borders (Borin and Mancini, 2015, 2019). In the example above, the car's components cross multiple borders before final assembly. Although this measure is not perfectly precise, it offers a tractable and intuitive way to identify GVC trade in the data.

GVC trade can be computed using readily available multi-regional input-output tables. These computations are illustrated schematically in Figure 2. The middle and right green trade flows, which cross at least two borders, are classified as GVC trade. At the bottom of the figure, Korea imports value-added from other countries and exports it to the United States, which subsequently exports it to the rest of the world (ROW). This value-added is counted as GVC trade, as it has crossed at least two borders. In contrast, non-GVC trade, defined as directly absorbed value-added exports (DAVAX), is represented by the leftmost orange line. It represents value added that originates in Korea, is exported to the United States, and absorbed in the U.S. market. Because this trade crosses only one border, it is

not classified as GVC trade.

The total value added embedded in Korea's exports, value added in exports (VAX), is defined as the sum of non-GVC and GVC exports originating from Korea, represented by the leftmost orange line and the middle green line. Gross exports (GX), represented by all three lines, capture the total trade flows from Korea to the United States and equal the sum of GVC trade and directly absorbed value-added exports (DAVAX). Accordingly, GVC trade can be calculated by subtracting DAVAX from GX.

Figure 2: GVC vs DAVAX

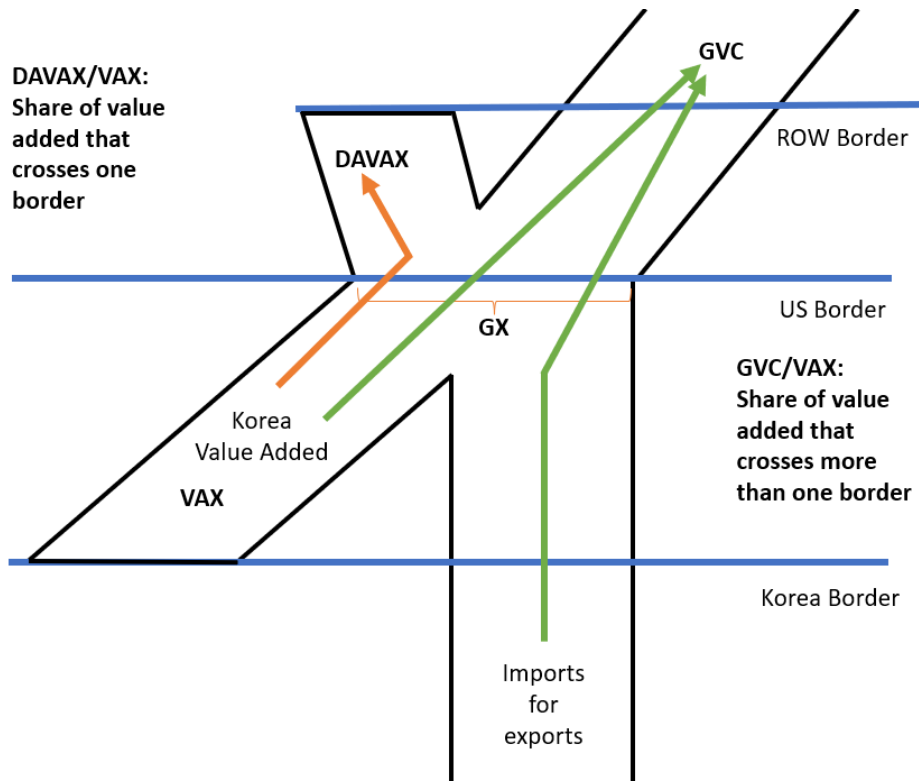
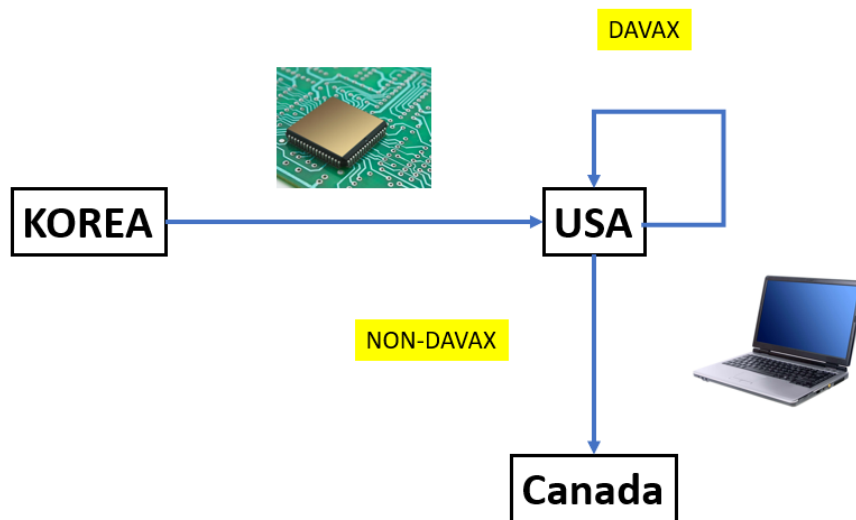


Figure 3 provides a real-world illustration of the framework shown in Figure 2. Korea exports a computer chip to the United States. If the United States uses the chip to produce a laptop that is sold to a domestic consumer, the Korean value added is classified as DAVAX, or non-GVC trade. However, if the United States exports the laptop to Canada, the Korean value added embodied in the chip is counted as GVC trade, since it crosses at least two borders. In the empirical analysis, the shares of GVC and non-GVC trade are

calculated by expressing DAVAX and GVC trade as proportions of total value added in exports (VAX).

Figure 3: DAVAX Example



4 Data

The data used to construct the GVC measures are drawn from Multi-Regional Input–Output (MRIO) tables, which are assembled from national input–output tables and international trade data by various organizations (Dietzenbacher et al., 2013; Timmer et al., 2015). This study uses the World Input–Output Database (WIOD) 2013 release for the period 1995–2006 and the Asian Development Bank (ADB) MRIO tables for 2007–2018. Two sources are used because WIOD extends only through 2014, while the ADB tables cover a longer time span, reaching 2021. The datasets are comparable, as the ADB tables are constructed using the WIOD framework and expanded to include additional countries and years. Combining both sources yields a panel of 40 countries and 35 industries from 1995 to 2018. The sample is restricted to 2018 to avoid data inconsistencies and to exclude the effects of the 2019 KORUS renegotiation.

Figure 4 presents a simplified two-country MRIO table. The unit of observation is a country–industry. Columns represent the inputs used by each country–industry, while

rows represent the inputs supplied by each country–industry. For instance, the first cell indicates that Korea’s agriculture sector uses 10 units of input from Korea’s agriculture sector itself. The cell below shows that it uses 15 units of input from Korea’s steel sector. Summing down a column aggregates all intermediate inputs and value added used by a given country–industry, which equals its total input use. Summing across a row aggregates all intermediate inputs supplied and final demand, which equals the sector’s gross output. For each country–industry, total inputs equal total gross output.

Figure 4: Input-Output Example

				1	2	3	4	5	6	7	8	9
				Input Use						Final Use		Total Use
				Korea	Korea	Korea	USA	USA	USA	Korea	USA	
				Agri	Steel	Auto	Agri	Steel	Auto	Household	Household	
1	Input Supplied	Korea	Agri	10	0	10	5	0	30	20	20	95
2		Korea	Steel	15	50	20	30	20	45	10	5	195
3		Korea	Auto	0	0	40	5	10	40	30	25	150
4		USA	Agri	10	0	0	20	0	0	20	40	90
5		USA	Steel	20	30	10	10	20	40	20	10	160
6		USA	Auto	0	5	55	10	30	60	10	55	225
7		Value added		40	110	15	10	80	10			
8		Gross output		95	195	150	90	160	225			

4.1 Calculation of GVC measures

This section provides algebraic definitions of three value chain concepts-GVC trade, non-GVC trade, and value-added trade-that are analyzed in the following sections. These definitions follow the notation used by Borin and Mancini (2019). Equation 1 defines gross exports from Korea to the US, GX_{KORUS} .

$$GX_{KORUS} = A_{KORUS}X_{US} + Y_{KORUS} \quad (1)$$

The variable GX_{KORUS} is an n-dimensional vector of exports, where n denotes the number of production sectors. The matrix A_{KORUS} is an n x n array of input coefficients. The $(i,j)^{th}$ element represents the demand for the i^{th} Korean input required per unit of output in the j^{th} US sector. In Equation 1, the first term captures Korean intermediate

goods exports to the United States, given by the product of the Korea–U.S. input coefficients matrix and U.S. output, while the second term represents Korean final goods exports to the United States. In the numerical example GX_{KORUS} can be calculated for the three sectors shown in rows 1-3 by adding the values in columns 4, 5, 6 and 8. These sums produce the elements of GX_{KORUS} equal to 55, 100, and 80.

Equation 2 defines non-GVC exports from Korea to the US, denoted $DAVAX_{KORUS}$. These exports are classified as non-GVC because they cross only a single border—the Korea–U.S. border. In Figure 2, $DAVAX_{KORUS}$ is represented by the leftmost trade arrow.

$$\begin{aligned} \mathbf{DAVAX}_{KORUS} &= \mathbf{V}_{KOR} (\mathbf{I} - \mathbf{A}_{KORKOR})^{-1} \mathbf{Y}_{KORUS} \\ &+ \mathbf{V}_{KOR} (\mathbf{I} - \mathbf{A}_{KORKOR})^{-1} \mathbf{A}_{KORUS} (\mathbf{I} - \mathbf{A}_{USUS})^{-1} \mathbf{Y}_{USUS} \end{aligned} \quad (2)$$

The first term in Equation 2 represents Korean value added embodied, either directly or indirectly, in final goods consumed by U.S. users. The second term similarly captures Korean value added embodied in intermediate goods exported to the United States that are subsequently transformed into final goods for U.S. users.

Equation 3 defines GVC trade as Korean exports to the United States that cross at least two borders before final absorption.

$$GVC_{KORUS} = GX_{KORUS} - DAVAX_{KORUS} \quad (3)$$

Given the definition of DAVAX as non-GVC exports, GVC exports can be obtained by subtracting DAVAX from total exports (GX). Importantly, the relationship between GVC trade and DAVAX is not fixed. For example, if GX increases by more than DAVAX, then GVC trade will also increase.

Finally, Equation 4 defines value-added exports (VAX).

$$\begin{aligned}
\mathbf{VAX}_{KORUS} = & \mathbf{V}_{KOR}(\mathbf{I} - \mathbf{A}_{KORKOR})^{-1} \left[\mathbf{Y}_{KORUS} + \mathbf{A}_{KORUS}(\mathbf{I} - \mathbf{A}_{USUS})^{-1} \mathbf{Y}_{USUS} \right. \\
& + \mathbf{A}_{KORUS}(\mathbf{I} - \mathbf{A}_{USUS})^{-1} \sum_{j \neq US, KOR}^G \mathbf{Y}_{USj} \\
& \left. + \mathbf{A}_{KORUS}(\mathbf{I} - \mathbf{A}_{USUS})^{-1} \sum_{j \neq US}^G \mathbf{A}_{USj} \sum_k^G \sum_{l \neq KOR}^G \mathbf{B}_{jk} \mathbf{Y}_{kl} \right]
\end{aligned} \tag{4}$$

In Equation 4, the first two terms inside the brackets, when multiplied by the expression preceding the brackets, sum to DAVAX, which is a component of VAX. The third and fourth terms capture Korean value added embodied in U.S. exports to other countries. Specifically, the third term measures U.S. final goods exports absorbed by other countries, while the fourth term captures Korean value added embodied in U.S. intermediate exports that are subsequently re-exported. Intuitively, Equation 4 calculates the total Korean value added embedded in exports to the United States that is ultimately absorbed outside Korea. Further details on the derivation of these expressions are provided in Borin and Mancini (2019).

VAX is used as the denominator when computing the share of GVC trade (GVC/VAX) and the share of non-GVC trade (DAVAX/VAX). Value-added exports are preferred to gross exports (GX) because GVC trade conceptually reflects value-added flows rather than gross trade flows.

5 Estimation Strategy

I employ the synthetic control methodology (Abadie et al., 2010; Arkhangelsky et al., 2019; Abadie, 2021) to estimate the treatment effect of KORUS on three outcome measures in the auto, agriculture, and electronics sectors.

1. Gross Exports (GX)
2. Share of GVC trade in value-added exports (GVC/VAX)
3. Share of non-GVC trade in valued-added exports (DAVAX/VAX)

Using time-series data on the three constructed outcome measures, I evaluate how they evolve relative to bilateral trade flows not affected by the provisions of the KORUS FTA.

The synthetic control method is a variation of the difference-in-differences approach. It constructs a synthetic counterfactual that closely resembles the treated unit by forming a weighted average of units in the control (donor) pool.

For U.S. exports to South Korea, the donor pool consists of U.S. exports to countries other than Korea and South Korean imports from countries other than the United States. For example, U.S. exports to Mexico and South Korean imports from Japan would be included. This restriction is imposed to identify bilateral trade pairs that most closely resemble U.S. exports to Korea. For U.S. imports from Korea, the donor pool includes U.S. import partners other than Korea and Korean export partners other than the United States. This construction applies across all sectors. In both the export and import specifications, country pairs that signed an FTA during the sample period are excluded to ensure that changes in the control group are not driven by other trade agreements.³

$$\tau_{1t} = Y_{1t}^T - Y_{1t}^C \quad (5)$$

τ_{1t} denotes the treatment effect of interest. In an ideal setting, both the treated outcome for GVC trade between the United States and Korea, Y_{1t}^T , and the untreated counterfactual outcome, Y_{1t}^C , would be observable, and the treatment effect would be obtained as their difference. In practice, however, only the treated outcome is observed. The synthetic control method addresses this problem by constructing an estimate of the unobserved counterfactual, that is, what GVC trade would have been in the absence of the FTA (Y_{1t}^C).

The synthetic control is constructed by selecting weights that minimize the difference between the pre-treatment characteristics of the treated unit and those of the control units. The objective is to match the treated unit's attributes using a weighted average of non-treated units. The pre-treatment characteristics included in the matching procedure are foreign value added (FVA), domestic value added (DVA), value added in exports (VAX), GDP, and several lags of the outcome variable. FVA measures value added generated outside the exporting country, while DVA measures value added generated domestically. FVA, DVA, and VAX are included to ensure that the synthetic control closely matches the scale and composition of value-added trade of the treated unit. GDP is included to match the economic size of the destination country.

Formally, let X_1 denote the $k \times 1$ vector of pre-treatment attributes for the treated

³The KORUS FTA may indirectly affect some bilateral pairs in the donor pool, potentially violating the independence assumption. However, its provisions are narrowly targeted toward U.S.–Korea trade flows and are unlikely to materially affect trade with third countries.

unit, and let X_0 denote the $k \times J$ matrix of corresponding attributes for the J control units. A $J \times 1$ vector of weights, W , is then chosen to minimize

$$\|X_1 - X_0 W\| \quad (6)$$

There are several approaches to minimizing this distance. I follow the method proposed by Abadie et al. (2010), which is analogous to minimizing a weighted sum of squared differences. A separate set of predictor weights, v_h , reflecting the relative predictive importance of each covariate, is determined using a data-driven procedure. The resulting weights are reported in the appendix.

$$\|X_1 - X_0 W\| = \left(\sum_{h=1}^k v_h (X_{h1} - w_2 X_{h2} - \dots - w_{J+1} X_{hJ+1})^2 \right)^{1/2} \quad (7)$$

The optimal weights w_j^* are then used to construct the counterfactual and obtain the estimated treatment effect $\hat{\tau}_{1t}$.

$$\hat{\tau}_{1t} = Y_{1t}^T - \sum_{j=2}^{J+1} w_j^* Y_{jt} \quad (8)$$

where Y_{1t}^T denotes the observed GVC outcomes for the treated unit and $\sum_{j=2}^{J+1} w_j^* Y_{jt}$ represents the weighted average of control units, which serves as an estimate of the counterfactual Y_{1t}^C . The difference between these two terms yields the estimated treatment effect of the KORUS FTA.

Inference in the synthetic control framework is conducted using placebo tests constructed by iteratively assigning the treatment to units in the donor pool. For example, in the case of U.S. exports to Korea, U.S. exports to China constitute one donor unit. In a given iteration, the same estimation procedure is applied as if the U.S.–China bilateral flow had received the treatment. Because this pair was not actually treated, its estimated treatment effect is expected to be small relative to the treated unit. Repeating this procedure for each donor unit generates a distribution of placebo treatment effects. If the estimated treatment effect for the treated unit is extreme relative to the distribution of placebo effects, it is considered statistically significant.

There are two approaches to determining extremeness. First, the gap between the observed treated outcome and its synthetic control can be plotted over time. If the post-treatment gap is substantially larger in absolute value than those observed for the placebo

units, the treatment effect is considered extreme. Visually, this corresponds to the treated unit’s post-treatment trajectory diverging more strongly than the placebo trajectories.

Second, significance can be assessed using the ratio of post-treatment to pre-treatment mean squared prediction error (MSPE). A large MSPE ratio indicates that the post-treatment fit deteriorates substantially relative to the pre-treatment fit. If the treated unit’s MSPE ratio is large compared to those of the placebo units, the treatment effect is considered extreme. A Fisher-type exact p-value can also be computed based on the rank of the treated unit’s MSPE ratio. Because inference in synthetic control settings is permutation-based, conventional 5% significance thresholds are not directly applicable. In this study, an MSPE ratio ranking among the top three is interpreted as significant.

6 Results

Table 4 reports the average treatment effects and indicates significance relative to the distribution of placebo effects. The results show that the KORUS FTA had significant effects on U.S. auto imports and U.S. agricultural exports, but no significant effects on the other trade flows examined.

U.S. auto imports increased by \$4.8 billion, driven by increases in both the GVC and non-GVC shares, suggesting that the FTA positively affected both types of trade in this sector. In contrast, the agreement reduced gross U.S. agricultural exports by \$0.7 billion. Although the share of non-GVC trade increased by 2.2 percentage points, the share of GVC trade declined more sharply—by 3.3 percentage points, resulting in a net negative effect. No significant impacts are detected for U.S. auto exports, agricultural imports, or electronics exports and imports.

Table 4: Table of KORUS FTA Treatment Effects

Sector	Outcome	US Exports to Korea		US Imports from Korea	
		Treatment Effect	Significant	Treatment Effect	Significant
Autos	GX	-\$0.3 billion	N	\$4.8 billion	Y
	GVC	5.6 pp	N	0.3 pp	Y
	DAVAX	-10.4 pp	Y	0.3 pp	N
Agri	GX	-\$0.7 billion	Y		
	GVC	-3.3 pp	Y		
	DAVAX	2.2 pp	Y		
Elec	GX	\$0.4 billion	N	-\$1.3 billion	N
	GVC	-1.2 pp	N	0.3 pp	N
	DAVAX	1.9 pp	N	0.3 pp	N

pp stands for percentage points

Across U.S. exports and imports in the three sectors, the heterogeneous effects reflect differences in production structures and value-chain integration. More regionally concentrated and deeply integrated industries, such as autos, exhibit increases in GVC-linked trade. In contrast, sectors with less integrated production structures, such as agriculture, display shifts toward non-GVC trade. The FTA appears to have no significant impact on electronics, a sector characterized by highly globalized and fragmented production networks. Taken together, these findings suggest that the effects of the FTA depend critically on sectoral production structures and the degree of embeddedness in global value chains. The following sections examine autos, agriculture, and electronics in greater detail, analyzing both U.S. exports to Korea and U.S. imports from Korea within each sector.

6.1 Autos

Figure 5 compares the three outcome measures for the treated unit, U.S. auto imports from Korea (solid blue line), with their corresponding synthetic controls (red dashed line) over the period 1998–2017. From left to right, the panels display U.S. gross auto imports from Korea, the GVC share of those imports, and the non-GVC share. The vertical line in 2012 marks the treatment year, when the KORUS FTA entered into force.

Figure 5a shows that gross U.S. auto imports from Korea increased substantially, rising from \$8.1 billion in 2009 to \$23.5 billion in 2015. This sharp increase underscores the importance of the FTA for Korea and may reflect both anticipatory and post-implementation effects. The estimated treatment effect, measured as the gap between the observed out-

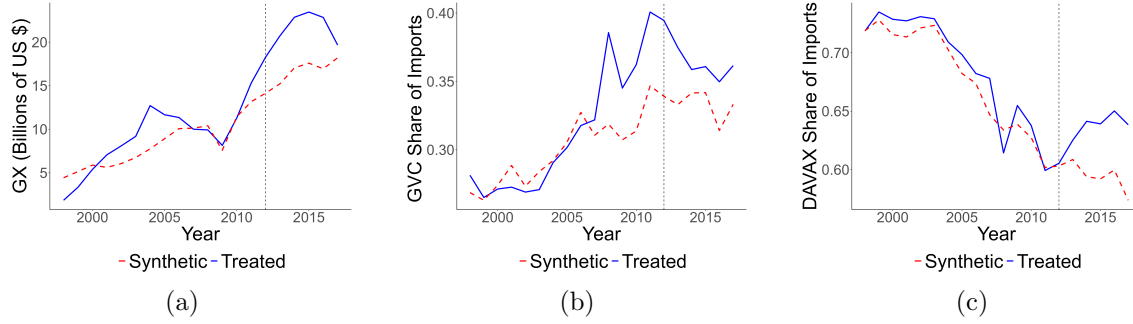
come (grey line) and the synthetic control (pink dashed line), averages approximately \$4.8 billion per year after 2012. Figures 5b and 5c indicate that both the GVC share and the non-GVC share of U.S. auto imports increased by 0.3 percentage points following the FTA when compared to the synthetic control, suggesting that the agreement positively affected both types of trade.

However, examining the trends in Figures 5b and 5c reveals a more nuanced pattern. Figure 5b shows that the share of GVC trade was trending upward prior to 2012 but shifts to a downward trend following the FTA's implementation. In contrast, Figure 5c indicates that while the agreement increased both the GVC and non-GVC shares, it appears to have altered the long-run composition of trade, making it increasingly less GVC-intensive after 2012.

One possible explanation is that the FTA disproportionately expands final goods trade relative to GVC trade, particularly given the substantial reduction in tariffs on finished autos. Although overall exports increased, much of this growth may have reflected Korean firms expanding direct exports of completed vehicles to the United States after 2012. In this scenario, GVC trade could increase in absolute terms while declining as a share of total trade. Another explanation is that the agreement encouraged greater localization of production through foreign direct investment in Korean firms' existing manufacturing facilities in the United States, such as the Kia assembly plant in Georgia and the Hyundai manufacturing plant in Alabama, thereby reducing the need for cross-border trade in intermediate inputs.

In either case, the outcome is beneficial for Korea, as it expands its automobile market share in the United States. For the United States, if the second explanation holds, the outcome may also be favorable, as increased foreign direct investment would strengthen domestic manufacturing capacity and generate spillover benefits for the local economy.

Figure 5: US Auto Imports from Korea vs Synthetic Control



Figures 6a and 6b show large positive differences for gross imports and the GVC share of trade relative to the placebos, suggesting significant treatment effects for both outcomes. However, these results may be partly driven by the general collapse in trade during 2008–2009. Lastly, no significant difference from the placebos is observed for the non-GVC share of trade (Figure 6c).

Figure 6: Placebo Test: US Auto Imports from Korea

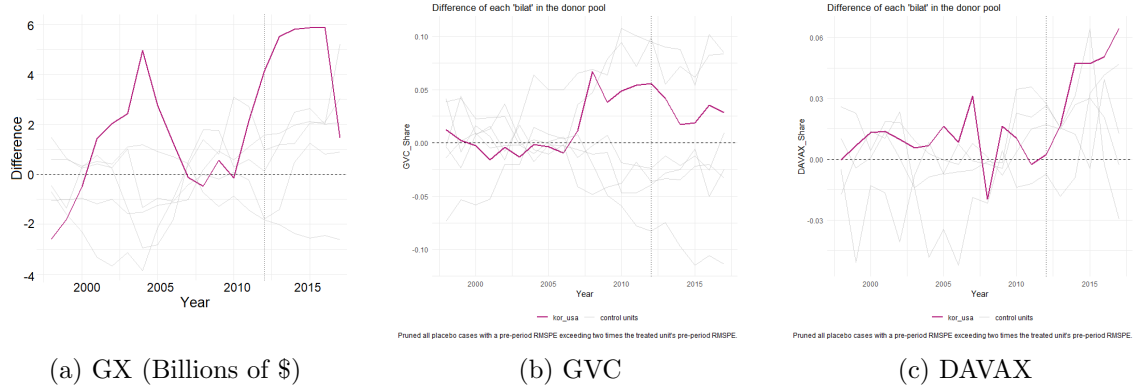


Table 5 shows that U.S. auto imports from Korea have the second-highest MSPE ratio for gross imports, providing additional evidence that the treatment effect is significant.

Table 5: Significance Table: US Auto Gross Imports from Korea

Trade Flow	MSPE Ratio	Rank
Sweden-US	6.35	1
Korea-US	5.19	2
Taiwan-US	3.06	3
France-US	1.97	4
China-US	1.64	5
UK-US	1.39	6
Japan-US	0.37	7
Canada-US	0.06	8

For U.S. auto exports to Korea, Figure 7a shows that gross exports initially increased from \$3.1 billion in 2011 to \$4.9 billion in 2012 and 2013. However, exports declined to \$3.4 billion in 2014. Although the average treatment effect on gross exports over 2012–2013 is -\$0.3 billion, the year-specific treatment effects were \$1.1 billion in 2012 and \$0.7 billion in 2013. The FTA has a much larger effect on the GVC share of trade, increasing it by 5.6 percentage points when compared to the synthetic control. In contrast to U.S. auto imports, the FTA has a large negative effect on the non-GVC share of trade, reducing it by 10.4 percentage points. In the case of U.S. auto exports, the FTA appears to have increased GVC trade while decreasing non-GVC trade.

However, once again, as with U.S. auto imports, the trends in Figures 7b and 7c shift in 2012. The share of GVC trade decreases steadily from 0.79 to 0.67, while the share of non-GVC trade increases from 0.57 to 0.71 after 2012. This pattern suggests that provisions targeting final goods generated stronger positive long-run effects, while those related to intermediate goods did not produce comparable gains. Due to this reversal in trend, the treatment effect is close to zero by 2017. One important caveat is that the sharp change in 2009, most likely driven by the Global Trade Collapse, is not captured by the synthetic control well. If the pre-2012 match during this period had been tighter, the estimated treatment effect may have been negative for the GVC share and positive for the non-GVC share.

However, considering U.S. auto exports to Korea as a whole, exports increase substantially over the period, rising from around \$2 billion to approximately \$5 billion by 2013. In addition, GVC intensity also increases markedly. This pattern suggests a bilateral re-

structuring of GVCs: while Korea appears to export more finished goods directly to U.S. consumers, the United States increasingly supplies intermediates and components used in Korean production.

Figure 7: US Auto Exports to Korea vs Synthetic Control

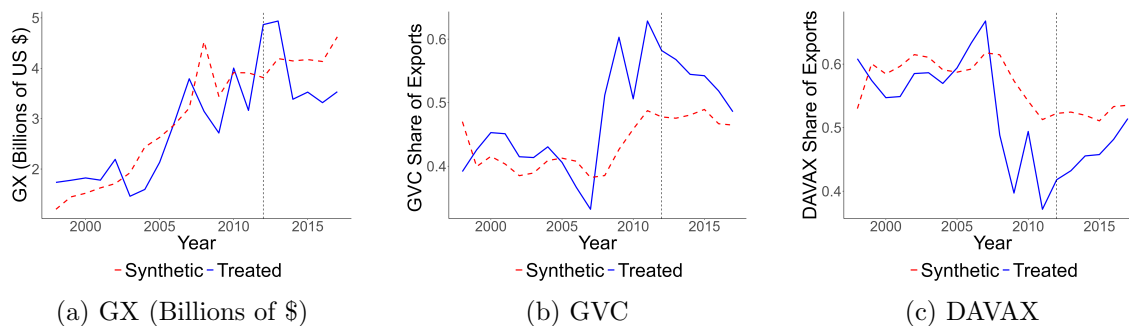
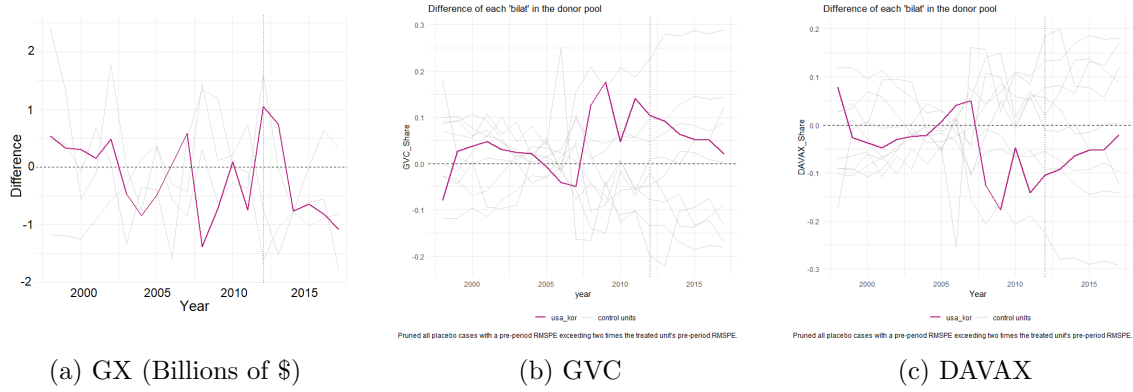


Figure 8 shows that only the share of non-GVC trade exhibits a large negative difference relative to the placebos, while the treatment effects of the FTA on gross exports and the share of GVC trade do not appear to be significant. The United States does not seem to have succeeded in expanding its presence in the South Korean auto market as a result of the KORUS FTA, despite some temporary gains in 2012–2013. What the FTA does appear to have accomplished is a reversal in the trend of U.S. exports to Korea, making them less GVC-intensive.

This mirrors earlier findings for U.S. auto imports. However, in this case, it is more likely that the FTA disproportionately expanded final goods trade relative to GVC trade, with American automakers increasing direct exports of completed vehicles to Korea rather than expanding localized production in U.S.-owned auto plants in Korea. Although the United States operates a GM plant in Korea, approximately 90 percent of its production is exported back to the United States. As a result, if production at this plant had increased, it would likely have raised the share of GVC trade rather than reduced it. Additional output would involve more cross-border production linkages, with value added generated in Korea embodied in vehicles that are subsequently shipped to the U.S. market rather than remaining in Korea and being absorbed domestically.

Figure 8: Placebo Test: US Auto Exports to Korea



6.2 Agriculture

As Figure 9 shows, the analysis of U.S. agricultural exports to Korea indicates a decrease in gross exports, a decline in the share of GVC trade, and an increase in the share of non-GVC trade. Contrary to expectations, the FTA had a negative treatment effect of \$0.7 billion on gross exports. This outcome likely reflects persistent consumer concerns in Korea regarding U.S. beef and mad cow disease, which prevented exports from increasing despite improved market access conditions.

Compared to auto exports, the share of GVC trade in agricultural exports is substantially lower (0.3 versus 0.8 in 2012). Because the FTA's provisions on agricultural products primarily targeted final goods, and agricultural products are typically consumed directly rather than re-exported, the agreement generated a negative treatment effect of 3.3 percentage points on the GVC share and a positive treatment effect of 2.2 percentage points on the non-GVC share. As with autos, the FTA appears to have shifted the composition of trade to be less GVC-intensive in the long-run.

Figure 9: US Agriculture Exports to Korea vs Synthetic Control

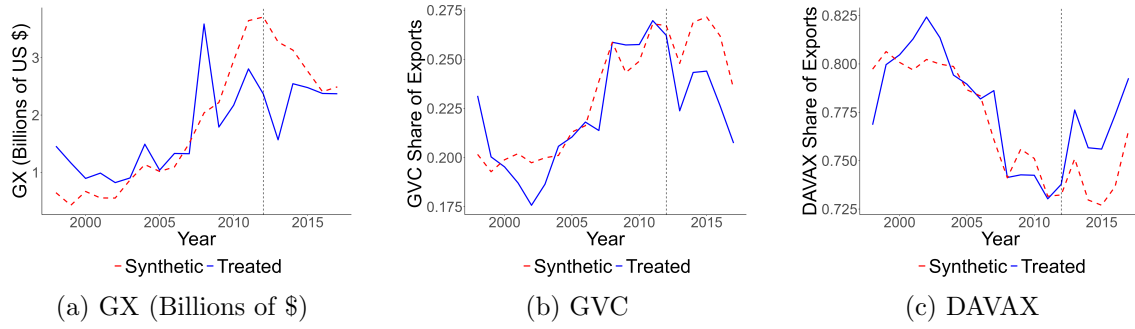
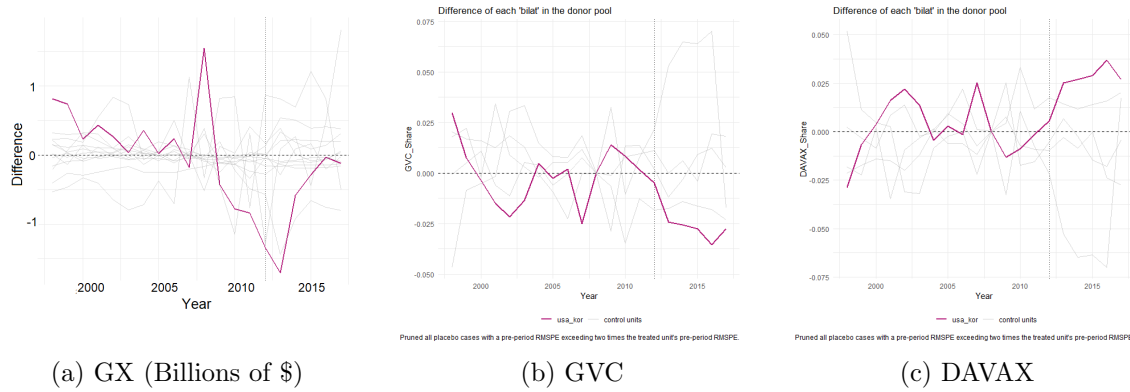


Figure 10 shows large differences for all three outcomes relative to their respective placebos, suggesting that the treatment effects are significant. Table 6 further indicates that U.S. agricultural exports to Korea have the highest MSPE ratio, reinforcing the evidence of a significant treatment effect ⁴

Figure 10: Placebo Test: US Agriculture Exports to Korea



⁴ A discussion of U.S. imports from Korea is omitted because the trade flow is economically insignificant.

Table 6: Significance Test: US Agricultural Exports to Korea

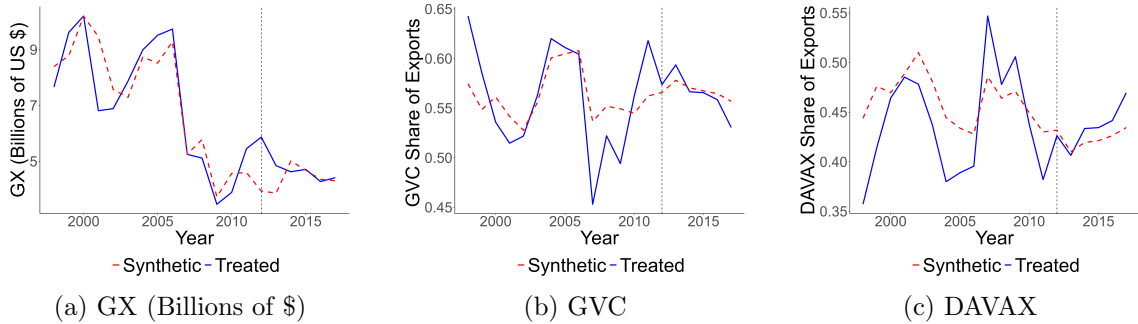
Trade Flow	MSPE Ratio	Rank
US-Korea	5.43	1
US-France	2.71	2
US-UK	2.44	3
US-Mexico	2.00	4
US-Spain	1.32	5
US-Italy	1.21	6
US-Turkey	1.15	7
US-Taiwan	0.96	8
US-Indonesia	0.82	9

6.3 Electronics

Table 4 indicates that there are no significant treatment effects in the electronics sector. This is consistent with expectations, as many trade barriers in electronics had already been removed prior to the FTA. Figure 11 shows that after 2012 the synthetic control closely tracks the observed outcomes of the treated unit across all three measures, implying that the estimated treatment effect is near zero.

These results provide a limited but useful validation of the synthetic control methodology. The approach detects no treatment effects in a sector where none would be anticipated. A similar pattern is observed for U.S. electronic imports. The corresponding figures and placebo tests for the electronics sector are presented in the appendix.

Figure 11: US Electronic Exports to Korea vs Synthetic Control



7 Gravity Estimation

This section provides a robustness check of the synthetic control results by estimating a gravity model to evaluate the effects of KORUS for each sector. The gravity dataset is constructed from the CEPII BACI data covering years 1995–2021, merged with the dependent variables derived from the synthetic control analysis (Gaulier and Zignago, 2010). It includes 62 countries and 35 sectors, consistent with the sample used in the synthetic control analysis.

For each dependent variable-GX, GVC, and DAVAX-two specifications are estimated. The first specification assumes a symmetric effect of KORUS on bilateral trade between Korea and the United States by using a binary indicator that takes the value of one when KORUS is in effect. The second allows for asymmetric effects by distinguishing the direction of trade flows, replacing the single KORUS indicator with two separate binary variables, one for each direction. In both specifications, a regional trade agreement (RTA) indicator is included, taking the value of one if the bilateral pair has a trade agreement in force, with the exception of KORUS. This controls for the effects of other trade agreements while separately identifying the impact of KORUS. The gravity equation is estimated using a multiplicative Poisson pseudo-maximum likelihood (PPML) estimator, which accommodates zero trade flows and includes exporter-year, importer-year, and bilateral pair fixed effects. Standard errors are clustered at the bilateral pair level.

Overall, the gravity results align with the broader narrative from the synthetic control analysis that trade became less GVC-intensive after KORUS. A plausible explanation is that the agreement’s provisions did not explicitly target the expansion of global value chain trade, and as a result, GVC trade did not expand as much as non-GVC trade. The broader implication is that free trade agreements alone do not necessarily foster deeper GVC integration without specific provisions aimed at facilitating such trade.

However, there are sector-specific differences between the results from the two methodologies. These differences arise for several reasons. First, the gravity regressions are estimated in levels rather than shares, as using levels is more appropriate in a structural gravity framework. Second, the synthetic control results depend on the composition of the donor pool, whereas the gravity estimates rely on fixed effects to control for multilateral resistance and other confounding factors. This comparison underscores the importance of methodological choice when evaluating free trade agreements. The remainder of this section examines the sector-specific results from the gravity estimations in greater detail.

Table 7 reports the gravity estimates for the auto sector. The results show no statistically significant effects for the RTA variable in any specification and no significant effect of KORUS under the symmetric specification. However, once the effects are allowed to differ by direction, KORUS becomes statistically significant, particularly for Korean auto exports to the United States. Gross Korean auto exports increase by approximately 48 percent following KORUS. GVC trade increases by about 45 percent, while non-GVC trade (DAVAX) rises by 51 percent, indicating a stronger expansion in non-GVC flows. In contrast, for U.S. auto exports to Korea, the only statistically significant effect is observed in GVC trade, which declines by roughly 49 percent.

Table 7: KORUS Effects in Auto Sector

	GX		GVC		DAVAX	
	(1)	(2)	(3)	(4)	(5)	(6)
	Sym.	Asym.	Sym.	Asym.	Sym.	Asym.
RTA	0.0464 (0.71)	0.0428 (0.65)	0.0505 (0.77)	0.0465 (0.70)	0.0540 (0.78)	0.0506 (0.72)
KORUS	0.216 (1.35)		0.162 (0.91)		0.249 (1.63)	
KORUS (US $\rightarrow$ KOR)		-0.367 (-1.66)		-0.399* (-2.02)		-0.342 (-1.32)
KORUS (KOR $\rightarrow$ US)		0.398*** (4.32)		0.372*** (3.80)		0.414*** (4.43)
Observations	72,313	72,313	72,313	72,313	72,313	72,313

t statistics in parentheses

Exporter-year FE, Importer-year FE, and Pair FE included.

Standard errors clustered by pair.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

These effects are economically large and broadly mirror the patterns observed in the synthetic control analysis. In both approaches, KORUS is associated with an increase in gross exports, with the expansion driven more by non-GVC trade than by GVC trade. This suggests that KORUS stimulated relatively “simpler” trade flows in both directions rather than deepening global value chain integration.

Turning to the agriculture sector, Table 8 shows no statistically significant effect of KORUS on U.S. agricultural exports to Korea, in contrast to the synthetic control results. While there is a statistically significant effect for Korean agricultural exports to the United States, the trade volumes are small and not particularly economically meaningful. These discrepancies raise questions about the reliability and sensitivity of the synthetic control method in this sector.

Table 8: KORUS Effects in Agriculture Sector

	GX		GVC		DAVAX	
	(1)	(2)	(3)	(4)	(5)	(6)
	Sym.	Asym.	Sym.	Asym.	Sym.	Asym.
RTA	0.0328	0.0342	0.0568	0.0573	0.00543	0.00732
	(0.57)	(0.60)	(0.82)	(0.82)	(0.09)	(0.13)
KORUS	-0.0223		-0.0690		-0.0160	
	(-0.14)		(-0.47)		(-0.10)	
KORUS (US → KOR)		0.00546		-0.0521		0.0147
		(0.03)		(-0.33)		(0.09)
KORUS (KOR → US)		-0.418**		-0.337*		-0.442**
		(-3.03)		(-2.41)		(-3.13)
Observations	72,174	72,174	72,174	72,174	72,174	72,174

t statistics in parentheses

Exporter-year FE, Importer-year FE, and Pair FE included.

Standard errors clustered by pair.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

In the electronics sector, the gravity results reported in Table 9 are statistically significant, particularly for U.S. exports to Korea, in contrast to the synthetic control estimates, which show no significant effects. Specifically, U.S. electronic exports to Korea decline by approximately 35 percent. This decline is driven mainly by GVC trade, which falls by about 55 percent, compared to a 17 percent decrease in non-GVC trade. For Korean electronic exports to the United States, the only statistically significant effect is observed in GVC trade, which decreases by roughly 23 percent. Again, the pattern mirrors the findings in the auto sector, with trade in both directions becoming less GVC-intensive

following KORUS.

Table 9: KORUS Effects in Electronics Sector

	GX		GVC		DAVAX	
	(1)	(2)	(3)	(4)	(5)	(6)
	Sym.	Asym.	Sym.	Asym.	Sym.	Asym.
RTA	0.0578 (1.52)	0.0584 (1.54)	0.0574 (1.37)	0.0583 (1.39)	0.0565 (1.38)	0.0567 (1.39)
KORUS	-0.206** (-2.74)		-0.285** (-2.99)		-0.137 (-1.85)	
KORUS (US → KOR)		-0.305*** (-5.78)		-0.440*** (-5.09)		-0.161** (-2.94)
KORUS (KOR → US)		-0.166 (-1.90)		-0.212* (-2.21)		-0.129 (-1.34)
Observations	71,494	71,494	71,494	71,494	71,494	71,494

t statistics in parentheses

Exporter-year FE, Importer-year FE, and Pair FE included.

Standard errors clustered by pair.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

These results underscore the importance of examining not only the asymmetric effects of free trade agreements but also sector-specific outcomes. They also highlight the significance of methodological choice in evaluating trade agreements. While the synthetic control method offers a transparent and intuitive framework for estimating treatment effects, it is subject to important limitations. In some sectors like the agriculture sector, the pre-treatment fit is weak, which raises concerns about the reliability of the constructed counterfactual and may explain why the results differ from those obtained using gravity estimates. Moreover, the synthetic control approach does not readily provide conventional statistical inference. Despite differences in sector-specific findings, the overarching pattern that emerges across both methodologies is consistent. KORUS appears to have made trade between Korea and the United States less GVC-intensive.

8 Conclusion

Theory predicts that the effects of FTAs on GVC and non-GVC trade depend on where liberalization occurs within the value chain. If liberalization focuses on final goods, non-GVC trade is expected to grow, while GVC trade remains unaffected or may weaken. This pattern is evident in the auto sector and, to a lesser degree, in the agriculture and electronics sectors. Evidence from KORUS suggests that while total trade increases, the composition shifts toward final goods and away from intermediate inputs, suggesting higher trade volumes without deeper integration of cross-border value chains.

The main objective of this paper is to empirically examine the effect of the KORUS FTA, one of the major U.S. trade agreements, on GVC and non-GVC trade. The effects of the agreement appear concentrated in sectors explicitly targeted during negotiations, notably autos and agriculture. The findings show that the agreement increased U.S. auto imports by an average of \$4.8 billion annually, while U.S. agricultural exports declined by \$0.7 billion per year. In autos, the shares of both GVC and non-GVC trade rose slightly, while in agriculture, the share of GVC trade declined and non-GVC trade expanded. The electronics sector was largely unaffected. However, across all three sectors, the composition of trade following the FTA suggests a shift toward non-GVC trade and away from GVC-intensive production.

Despite being the larger partner and having greater leverage in shaping the terms and sectoral coverage of the agreement, South Korea appears to have benefited more, while the United States realized only limited gains. Expanding U.S. auto and agricultural exports was one of the stated objectives of the agreement. However, U.S. auto exports rose only briefly before returning to pre-agreement levels, and agricultural exports declined modestly. In contrast, Korea significantly expanded its auto exports to the United States. Moreover, Korea preserved its important political alliance with the United States dating back to the Korean War.

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9 Appendix

Figure 12: Weights used to construct Synthetic Control for US Auto Exports to Korea

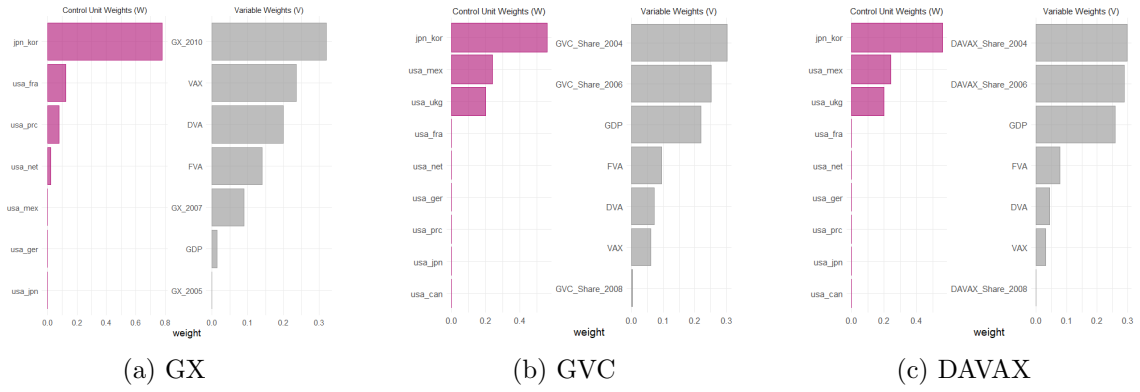


Figure 13: Weights used to construct Synthetic Control for US Auto Imports from Korea

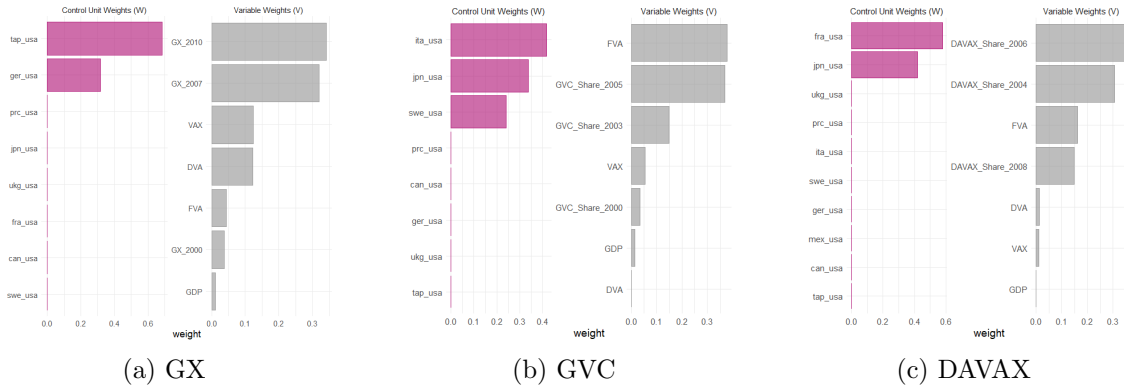


Figure 14: Weights used to construct Synthetic Control for US Agricultural Exports to Korea

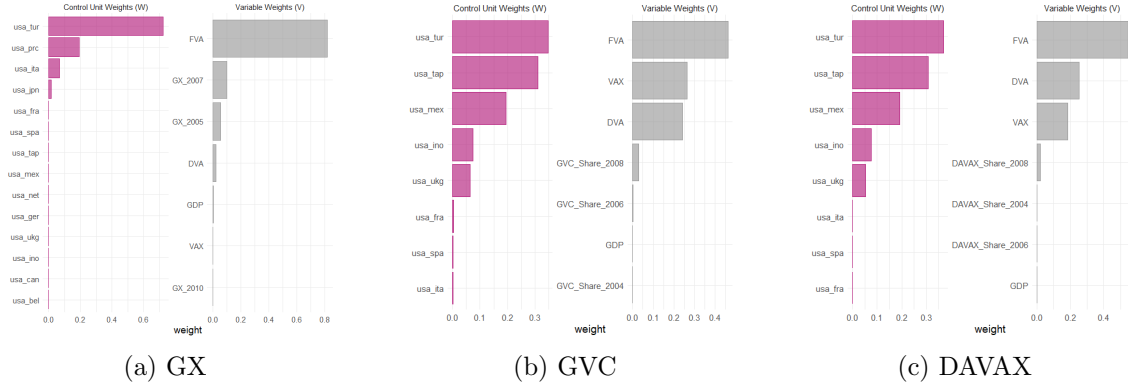


Figure 15: Weights used to construct Synthetic Control for US Agriculture Imports from Korea

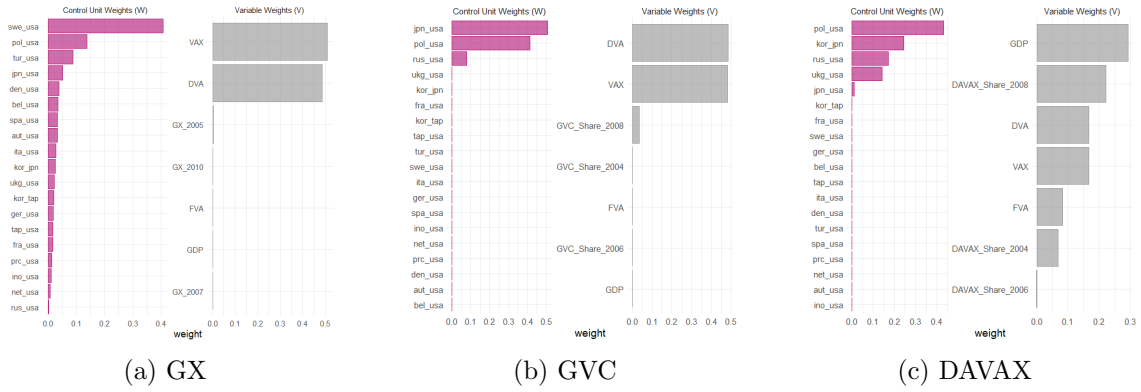


Figure 16: Weights used to construct Synthetic Control for US Electronic Exports to Korea

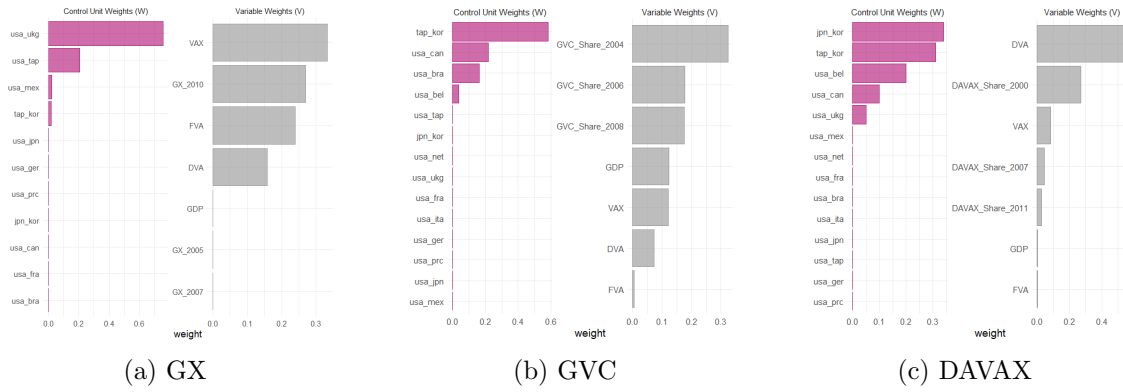


Figure 17: Weights used to construct Synthetic Control for US Electronic Imports from Korea

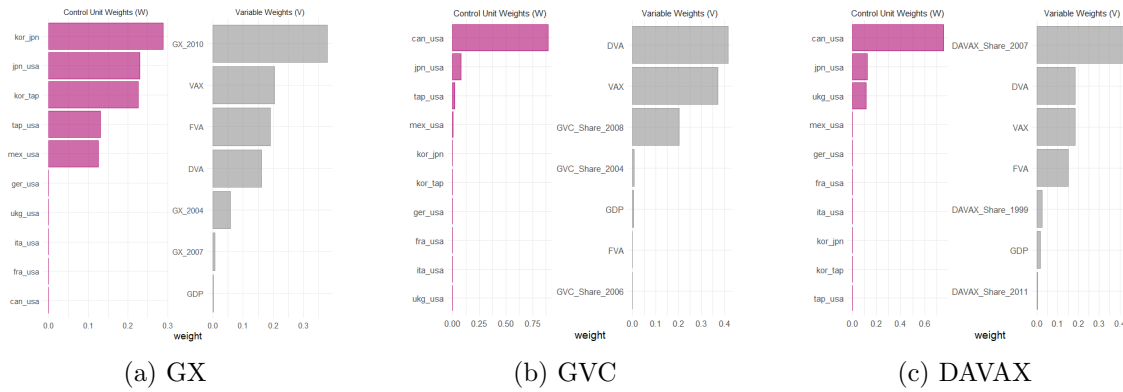


Figure 18: US Electronic Imports from Korea vs Synthetic Controls

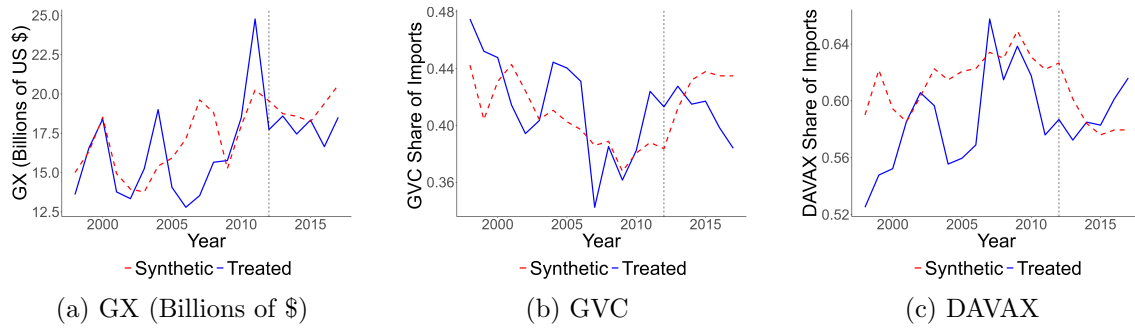


Figure 19: Placebo Test: US Electronic Imports from Korea

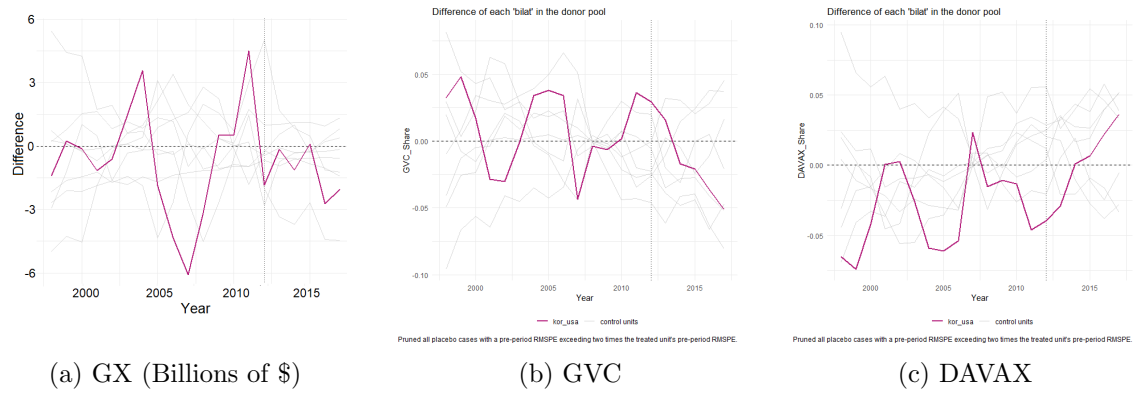


Figure 20: Placebo Test: US Electronic Exports to Korea

